

Permutation groups — homework problems

Peter Zeman

Submission instructions. To submit homework problems, create an Overleaf project with the title `2026_your_name_pg`. Then provide me the editing rights to the project. To pass the course it is required to solve **all problems** in this document **by the end of the academic year**.

Problem 1. Recall that a simple graph is a tuple (V, E) , where V is a finite set and $E \subseteq \binom{V}{2}$. A digraph is a tuple (V, E) , where $E \subseteq V^2$.

- (1) Construct a simple graph X such that $\text{Aut}(X)$ is isomorphic to the cyclic group $\mathbb{Z}_n$.
- (2) Let $G = \{g_1, \dots, g_n\}$ be a group. Define a colored digraph X with $V(X) = G$ such that there is a directed edge of color k from g_i to g_j if

$$g_i g_j^{-1} = g_k \iff g_i = g_k g_j.$$

What is $\text{Aut}(X)$? Note that an automorphism of a colored digraph has to preserve the colors and also the orientations of the edges.

- (3) (Frucht's theorem) Given a finite group G , there exists a simple graph X with $\text{Aut}(X) \cong G$.